

ACME Bearing Retainer — Optimizing Manufacturing Flow Through Value Stream Mapping

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Mapped the current state of ACME's bearing retainer manufacturing process — identifying 14-day lead time driven by inventory waste and push scheduling, with value-add cycle time of just 60 seconds. Future state recommendations target 2–3 day lead time through kanban and daily milk-run deliveries.

TOOLS & METHODS

Value Stream Mapping · Takt Time · Bottleneck Analysis · DOWNTIME Waste Framework · SMED · Kanban
/ Pull System Design · Lead Time Calculation

Executive Summary

ACME Incorporated's bearing retainer manufacturing line in Guelph, Ontario was producing to a 14-day production lead time while the actual value-add cycle time was only 60 seconds — a process efficiency of less than 0.01%. This VSM project mapped the current state in full, calculated all key lean metrics (Takt time, bottleneck, lead time, cycle time, process efficiency), applied the DOWNTIME waste framework, and designed a future state to reduce lead time from 14 days to 2–3 days while decreasing WIP inventory by approximately 79%. Recommended improvements included Kanban implementation, increased delivery frequency, SMED initiatives, and line balancing.

Business Problem

Company	ACME Incorporated, Guelph, Ontario
Industry	Automotive Parts Manufacturing — Tier 2 Supplier
Product	Large (L) and Small (S) Bearing Retainers
Customer	ABC Enterprises — daily shipments in returnable trays (60 retainers/tray)
Supplier	XYZ Steel Company — weekly raw sheet delivery (Mondays only)
Problem	14-day production lead time with only 60 seconds of value-add work. Push scheduling creating 4,800 pieces of WIP. Weekly batch raw material purchasing locking in 5 days of inventory. Scheduling system not responsive to actual customer demand.
Scheduling	Current: weekly push forecast — not demand-driven

Objectives

- Map the complete current state value stream from raw material supplier to customer delivery.
- Calculate Takt time, bottleneck, capacity, production lead time, and cycle time.
- Identify and categorise all waste using the DOWNTIME framework.
- Design a future state with quantified lead time reduction targets and specific lean interventions.

Process Analysis

Key Calculations

Metric	Calculation	Result
Takt Time	Available time ÷ Customer demand = 25,200 sec/day ÷ 720 units/day	35 seconds/unit
Bottleneck	Operation with longest cycle time Press: 3s Pierce & Form: 22s Finish Grind: 35s	Finish Grind (35s)
Daily Capacity	Available time ÷ Bottleneck CT = 25,200 ÷ 35 = 720 units	720 units/day
Production Lead Time	Sum of all queue/wait times between operations = 5d (RM) + 1.67d + 1.25d + 3.75d + 2.30d (ship)	~14 days
Total Cycle Time	Sum of all value-add processing times = 3s (Press) + 22s (Pierce & Form) + 35s (Grind)	60 seconds
Process Efficiency	Value-add time ÷ Total lead time = 60s ÷ (14 × 86,400s)	<0.01%

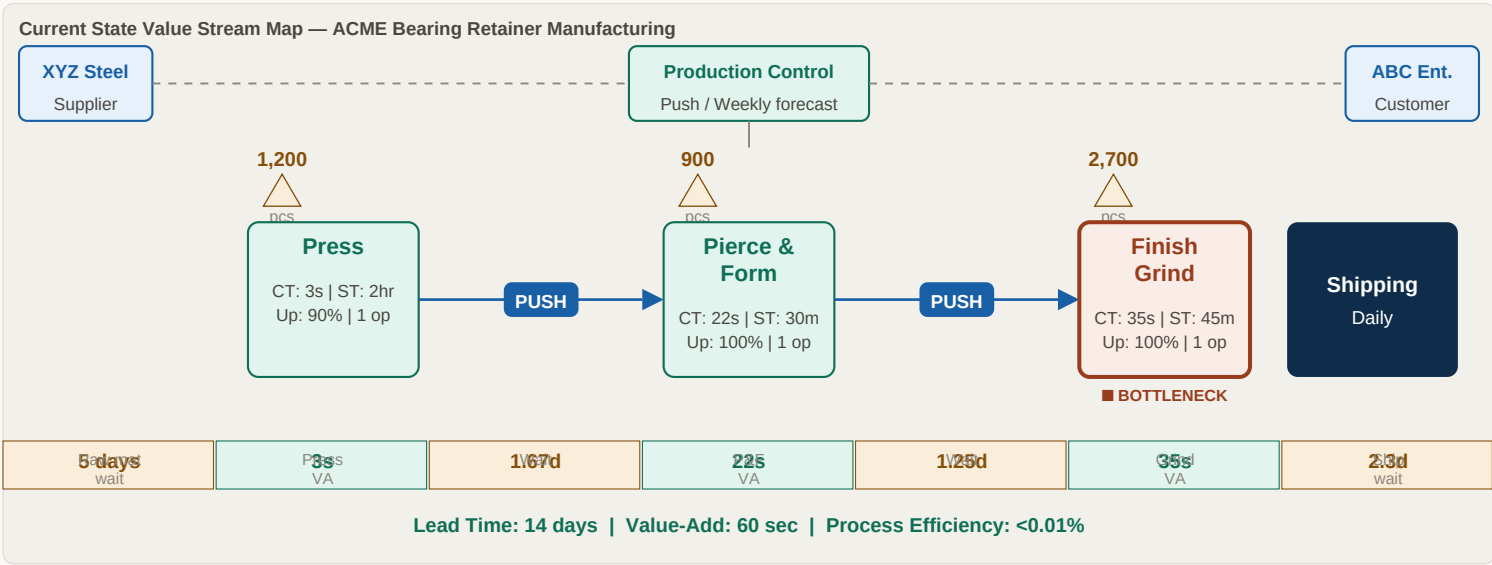
Inventory Queue Breakdown

Queue Location	WIP Quantity	Days of Inventory	% of Lead Time
Raw material (before Press)	1,200 pcs	1.67 days	12%
Before Pierce & Form	900 pcs	1.25 days	9%
Before Finish Grind	2,700 pcs	3.75 days	27%
Raw material delivery wait	—	5.00 days	36%
Finished goods / shipping	—	2.30 days	16%
TOTAL	4,800 pcs WIP	13.97 days	100%

Analysis & Key Findings

35 sec Takt Time	14 days Production Lead Time	60 sec Value-Add Cycle Time	<0.01% Process Efficiency
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Current State Value Stream Map



The 14-day lead time is composed almost entirely of inventory wait time — the product spends 99.99% of its time sitting in queues. The 5-day raw material wait (caused by weekly batch delivery) is the single largest contributor at 36% of lead time, yet it is entirely a supply chain scheduling decision — not a physical process constraint.

DOWNTIME Waste Identification

DOWNTIME Waste Analysis		
I	Critical	Critical — 4,800 WIP pieces in queue
O	High	High — Push scheduling drives excess inventory
W	High	High — 14 days lead time vs 60 sec value-add
T	Medium	Medium — Batched inter-op movement
M	Medium	Medium — Press setup 2hr; materials not co-located
D	Medium	Medium — Rework loop implied; no explicit rate
E	Medium	Medium — Push scheduling admin overhead
N	Low	Low — Single-machine operators; cross-train opportunity

Inventory (I) is the dominant waste — 4,800 WIP pieces visible in queue, with Finish Grind backlog alone (2,700 pcs) representing 3.75 days of waiting. Overproduction and Waiting wastes are directly caused by the weekly push scheduling system, which produces to forecast rather than actual customer demand.

Recommendations — Future State Design

Priority	Lean Intervention	Expected Benefit
01 — High	Implement Kanban / Pull System Replace weekly push forecast with kanban cards triggered by ABC Enterprises actual daily orders. Eliminates batch overproduction and excess WIP.	Lead time ↓ 60% WIP ↓ 60–70%
02 — High	Increase Raw Material Delivery Frequency Negotiate with XYZ Steel: shift from weekly batch to bi-weekly or daily milk-run deliveries. Removes 5-day RM inventory buffer — the single largest lead time contributor.	RM Inventory ↓ 70% Lead time ↓ 36%
03 — Medium	SMED on Press Station (2hr → <30min) Reduce setup from 2 hours to <30 min using SMED: externalise prep tasks, standardise tooling, parallel setup. Enables smaller economic lot sizes and greater scheduling flexibility.	Setup time ↓ 75% Smaller lot sizes
04 — Medium	Balance Line to Takt Time Press at 3s and Pierce & Form at 22s are under-utilised vs Finish Grind at 35s. Assign Press operator multi-process handling during available time.	Throughput ↑ 15%
05 — Ongoing	Future State Lead Time Target With kanban + milk-run + balanced line: lead time from 14 days to 2–3 days. Process efficiency improves from <0.01% to ~0.05%.	LT: 14d → 2–3d VA ratio 5× better
06 — Ongoing	Continuous Improvement Cadence Weekly VSM review; KPI dashboard monitoring CT, lead time, WIP, and OEE on Finish Grind. Quarterly kaizen events to sustain gains.	OEE sustained Kaizen culture

Financial Impact & Expected Benefits

14d→2-3 d	4,800→~ 1,000	5×	720→72 0+
Lead Time Target	WIP Pieces	Process Efficiency Gain	Capacity u/day

Metric	Current State	Future State	Improvement
Takt time	35 sec (matched to demand)	35 sec	Maintained — no change needed
Production lead time	~14 days	~2–3 days	~80% reduction
WIP inventory	4,800 pieces	~1,000 pieces	~79% WIP reduction
RM inventory	5 days (weekly delivery)	1–2 days (daily milk-run)	70% RM inventory reduction
Process efficiency	<0.01%	~0.05%	5× improvement
Scheduling	Push (weekly forecast)	Pull (kanban — demand-driven)	Eliminates overproduction waste
Press setup time	2 hours	<30 minutes (SMED)	75% setup time reduction

- Reducing WIP from 4,800 to 1,000 pieces releases floor space and working capital — fewer pieces in queue means faster defect detection and lower cost of quality.
- A 2–3 day lead time vs. 14 days is a competitive advantage: it enables ACME to respond to short-notice customer demand changes without holding excess finished goods inventory.
- The SMED improvement on the Press station enables smaller lot sizes — reducing inventory buildup between Press and Pierce & Form, and making the line more responsive to the mix of Large vs. Small bearing retainer orders.

Key Insights & Future Opportunities

Key Insights

- **Process efficiency is a revealing metric:** every team knew lead time was long, but framing it as <0.01% process efficiency — the product spends 99.99% of its time waiting — made the waste viscerally clear in a way that raw days figures did not.
- **The bottleneck is not always the problem:** Finish Grind is the bottleneck (35s CT) — but its cycle time exactly matches Takt time. The capacity is correct. The problem is not the bottleneck process; it is the WIP accumulated in front of it due to push scheduling.
- **Supply chain scheduling is a lean lever:** the single largest reduction in lead time comes from changing the raw material delivery frequency — a supply chain negotiation, not a process engineering change. Lean scope extends beyond the factory floor.
- **SMED changes what is economically possible:** a 2-hour setup on the Press forces large lot sizes. Reducing setup to 30 minutes enables smaller lots, which then reduces WIP queue naturally — without any additional constraints.

Future Opportunities

- Draw the Future State VSM with kanban loops, supermarket pull points, and daily milk-run delivery icon — making the target condition concrete for the implementation team.
- Implement OEE tracking on Finish Grind (the bottleneck) — any downtime on the bottleneck directly reduces system output, making it the highest-priority equipment for TPM and availability improvement.
- Conduct a kaizen event targeting the Finish Grind setup time (45 min) using SMED to further increase scheduling flexibility at the system constraint.

Core Competencies Applied

Skill / Competency	Evidence in This Project
Value Stream Mapping	Current state VSM constructed with full process data, inventory queues, push flows, and timeline
Takt Time Calculation	$\text{Takt} = 25,200\text{s} \div 720 \text{ units} = 35\text{s}$; matched to customer demand, compared to bottleneck CT
Bottleneck Analysis	Finish Grind identified as bottleneck (longest CT); capacity constrained to 720 units/day
Lead Time Analysis	14-day lead time decomposed into 5 segments; 5-day RM wait = 36% — highest-leverage target

DOWNTIME Waste Framework	All 8 DOWNTIME waste categories identified and severity-rated against current state data
Lean Tool Selection	Kanban, milk-run, SMED, line balancing — each matched to specific waste type identified in VSM